

Adversarial Dec-POMDP Game Theory in Decentralized Multi-Agent Pursuit–Evasion: Mitigating Freeze-Loops and Information Asymmetry

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zero-trust-cop

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Abstract—We report a failure mode observed in a live, cryptographically audited pursuit–evasion league between mutually distrusting reinforcement-learning agents. The pursuer never observes the evader’s position; it maintains a belief from a *self-reported* pheromone signal the evader is free to fabricate. We prove that this construction admits a policy fixed point in which the pursuer becomes permanently stationary, and we observed it in a graded match: the pursuer emitted **STAY** for exactly sixteen consecutive turns in each of three sub-games, 52% of all its turns. We give the closed form of the belief field, derive the switching lag L between a signal change and the belief following it, and validate that expression against simulation to the turn. We then evaluate a minimal mitigation—a refutation rule keyed on the pursuer standing on its own belief with no capture to claim—over 1,000 simulated games across four evader profiles. The mitigation eliminates the freeze entirely (longest stationary run $23 \rightarrow 0$) and, we report, *does not convert it into captures*: a grid search over 80 hyperparameter configurations and 16,000 further games yields zero captures against a perfectly-informed deceptive evader. We argue the residual gap is a train/test mismatch rather than a heuristic one, and that the signalling channel is cheap talk whose only sequential equilibrium is babbling.

Index Terms—Dec-POMDP, pursuit–evasion, cheap talk, babbling equilibrium, belief state, multi-agent systems, commit–reveal

I. INTRODUCTION AND THE PURSUIT–EVASION SETUP

Two independent agents—a pursuer p and an evader e —play a simultaneous-move game on a grid $G \subset \mathbb{Z}^2$ of size 7×7 over a horizon $T = 35$. They share no process, no memory and no trust. Every move is sealed by a SHA-256 commit–reveal exchange before either engine advances, and each sub-game closes with a mutual audit in which each peer re-hashes the other’s disclosed chain.

The information structure is the paper’s subject. Neither agent observes the other’s position. Instead each transmits, once per turn, a *scent grid* $\sigma_t : G \rightarrow \mathbb{R}_{\geq 0}$, and each maintains a belief field Φ_t updated from what it receives. Crucially σ_t is **self-reported and unverified**: nothing in the protocol binds it to the sender’s true position, and the sealed commitment covers the move record, not the grid.

II. THE BABBLING EQUILIBRIUM AND THE INFINITE STAY LOOP

A. Notation

Let $p_t, e_t \in G$. The action set is $A(p_t) = \{p_t + \delta \in G \mid \delta \in \Delta\}$ with $\Delta = \{(0, 0)\} \cup \{\delta : \|\delta\|_1 = 1\}$, where $(0, 0)$ is **STAY**. The pursuer’s belief is $b_t = \operatorname{argmax}_{g \in G} \Phi_t(g)$ and its policy is

$$a_t = \pi(p_t, b_t) = \operatorname{argmin}_{a \in A(p_t)} \|b_t - a\|_1, \quad (1)$$

ties broken by a learned action-value table. The field evolves as

$$\Phi_{t+1} = (1 - d) \Phi_t + K \left(\operatorname{argmax}_g \sigma_t \right), \quad (2)$$

with decay $d \in (0, 1)$ and $K(c)$ a Manhattan-falloff kernel centred at c .

B. STAY as a unique optimum

Proposition 1. *STAY is uniquely optimal under (1) if and only if $b_t = p_t$.*

Proof. G is convex under the ℓ_1 metric, so if $b_t \neq p_t$ there exists $a \in A(p_t) \setminus \{p_t\}$ with $\|b_t - a\|_1 = \|b_t - p_t\|_1 - 1$, strictly better than $(0, 0)$. Conversely if $b_t = p_t$ then $\|b_t - p_t\|_1 = 0$ while every other legal action yields 1. $\square$

This is the critical structural point: the learned table cannot override the result. Our policy admits it only as a tie-break *within* the distance-optimal set, and when $b_t = p_t$ that set is the singleton $\{\text{STAY}\}$, whatever the learned values.

C. The fixed point

Theorem 1 (Freeze). *If $b_t = p_t$ and the evader signals a constant cell $c = \operatorname{argmax} \sigma_\tau = p_t$ for all $\tau \geq t$, then $p_\tau = p_t$ for all $\tau \geq t$.*

Proof. By induction. The base case is $\Phi_t(p_t) > \Phi_t(g)$ for $g \neq p_t$, by uniqueness of the argmax. Assume $\Phi_\tau(p_t) > \Phi_\tau(g)$. Since $1 - d > 0$ and $K(p_t)(p_t) > K(p_t)(g)$,

$$\Phi_{\tau+1}(p_t) = (1 - d)\Phi_\tau(p_t) + K(p_t)(p_t) \quad (3)$$

$$> (1 - d)\Phi_\tau(g) + K(p_t)(g) = \Phi_{\tau+1}(g). \quad (4)$$

Hence $b_{\tau+1} = p_t$, and by the Proposition π selects $p_{\tau+1} = p_\tau$. The pursuer’s trajectory is stationary. $\square$

D. Belief dynamics and the switching lag

Under repeated deposit at c_0 from $\Phi_0 = 0$,

$$\Phi_t(g) = \frac{1 - (1 - d)^t}{d} K(c_0)(g), \quad (5)$$

whose steady state is $K(c_0)(g)/d$. If the signal then shifts to c_1 , let $M_0 = \Phi_{\text{pre}}(c_0) - \Phi_{\text{pre}}(c_1)$ be the accumulated mass difference and $\Delta K = K(c_1)(c_1) - K(c_1)(c_0)$. The belief follows the signal only after

$$L = \left\lceil \frac{\ln\left(1 + \frac{d M_0}{\Delta K}\right)}{-\ln(1 - d)} \right\rceil \quad (6)$$

further observations. We validated (6) against the implementation at $d = 0.1$: for a signal parked for 6, 12 and 18 turns the predicted lags were 4, 6 and 6, and the measured lags were 4, 6 and 6.

E. Cheap talk

The signal enters no utility and no transition: $u_i(p_{1:T}, e_{1:T}, \sigma_{1:T}) = u_i(p_{1:T}, e_{1:T})$ for both players. It is therefore *cheap talk*. In a strictly adversarial game the players’ interests are opposed, and the only sequential equilibrium under Perfect Bayesian Equilibrium is the **babbling equilibrium**: the sender transmits uninformative signals and a rational receiver assigns them no credibility. Our pursuer is not rational in that sense—its update rule is fixed—so the evader faces a single-controller problem and optimally chooses σ_t to steer b_t . The Freeze theorem gives its best response in closed form: pin $\arg\max \sigma$ to a corner and evade elsewhere.

III. THE THAWED PURSUER

We adopt the strongest refutation the protocol affords. Capture is settled by claim and honest answer, so a pursuer standing on the evader’s cell would have a capture to claim; standing there with none proves the evader is elsewhere. Formally, STAY is removed from $A(p_t)$ whenever $b_t = p_t$ and no capture is claimable, and additionally whenever the stationary run exceeds a configured bound κ . The learned table is untouched: the restriction applies to the action set *before* either ranking stage, never after.

Note this is an *empirical* refutation, requiring no model of the opponent’s honesty and no additional message—which is why we prefer it to filtering σ_t for physical plausibility.

IV. EXPERIMENTAL RESULTS

A. Method

Because the training harness supplies true positions, it cannot reproduce the fault; we therefore built a harness that reproduces the wire, in which the evader’s transmitted grid and its true cell are independent. Four profiles: (A) random walker, honest; (B) greedy distance-maximiser, honest; (C) deceptive bluffer—evades while pinning $\arg\max \sigma$ to a corner; (D) corner-hider, honest. 125 games per profile per arm.

B. Results

TABLE I
THAWED VERSUS UNTHAWED, 1,000 GAMES

Profile	Arm	Capture	Longest STAY	Thaw fired
A random	thawed	100.0%	0	171
A random	unthawed	99.2%	4	—
B greedy	thawed	0.0%	0	500
B greedy	unthawed	0.0%	6	—
C bluffer	thawed	0.0%	0	1500
C bluffer	unthawed	0.0%	23	—
D corner	thawed	100.0%	0	0
D corner	unthawed	100.0%	0	—

C. A negative result

The freeze is eliminated: the longest stationary run against the bluffer falls from 23 to 0. It does not become capture. Profiles B and C are 0% in *both* arms, so the freeze was never the reason a competent evader escapes. We further searched 80 configurations of decay $d \in [0.05, 0.25]$, emission intensity $\in [0.5, 1.0]$ and bound $\kappa \in \{2, 3, 4, 5\}$ at 200 games each—16,000 games—against profile C. **No configuration captured anything**, and no knob moved any metric. We report this rather than crown a least-bad cell.

Two limitations bound the claim. Profiles B and C are given the pursuer’s true position while the pursuer has only belief, so 0% means “against a perfectly-informed evader”, an upper bound on evader strength. And the shipped contract has a null barrier seed, so every game runs on an empty board.

V. PROTOCOL: UNIFIED ENDPOINT AND COMMIT–REVEAL

Each half-turn carries $\text{commit} = \text{SHA256}(\text{canonical}(r) \parallel \nu)$ over the sealed record $r = \{\text{step, state, position, move, intent, hint}\}$ with a fresh nonce ν ; records and nonces are disclosed only at the closing audit, where the opponent re-hashes each against the digest pushed at the time. Self-consistency alone would be a checksum—a peer rewriting its whole chain passes it—so it is the comparison against the *pushed* digest that makes the seal binding.

Serving both roles from one endpoint removes a class of operator error, but introduces one: a peer may reach itself. The mispairing check must therefore compare the caller’s declared role against an *authoritative* local role. Deriving the local role from the caller’s claim makes the comparison tautological and destroys the only place a mispairing is caught—a mispairing

both engines otherwise play through coherently for the full horizon, producing artifacts that join perfectly and a contradiction visible only to a human reader.

VI. CONCLUSION AND FUTURE WORK

An unverified belief channel in an adversarial Dec-POMDP admits a policy fixed point that a distance-minimising pursuer cannot escape, and we observed it costing 52% of a graded match. The refutation rule removes it completely and is not, by itself, worth a single capture against a competent evader.

The residual gap is a train/test mismatch: the action-value tables were trained against *true* opponent positions and are executed against an argmax that lags by L turns per (6). Future work is to retrain on belief-derived state so that training and execution share a state distribution, and to evaluate against evaders that are themselves belief-limited rather than perfectly informed.

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